

PERTH METRO, Australia

WMO: 946080

Lat: 31.9192S

Lon: 115.8728E

Elev: 25

StdP: 101.02

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.9	5.1	0.6	4.0	17.7	1.9	4.3	15.3	8.4	15.9	7.4	15.4	0.4	50	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.9	36.4	20.2	34.5	19.8	32.6	19.5	22.7	30.1	21.7	29.6	21.0	28.6	3.9	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.5	15.2	25.4	19.5	14.3	24.4	18.7	13.5	23.8	67.1	29.5	63.5	29.7	60.8	28.6	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.7	6.1	1.5	40.5	0.9	1.7	0.8	41.7	0.3	42.7	-0.2	43.6	-0.9	44.8		
(5)			1.4	24.1	0.9	1.2	0.7	24.9	0.2	25.6	-0.3	26.2	-1.0	27.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.5	24.3	24.5	22.8	19.6	16.2	13.8	13.0	13.5	14.6	17.0	20.2	22.5		
	DBStd	5.03	3.27	3.27	3.30	2.91	2.47	2.14	2.25	2.09	2.28	2.86	3.22	3.50		
	HDD10.0	4	0	0	0	0	0	1	3	1	0	0	0	0		
	HDD18.3	738	1	1	3	19	73	136	164	151	113	60	14	3		
	CDD10.0	3098	444	405	397	287	193	115	97	109	139	218	306	386		
	CDD18.3	789	187	172	142	56	8	1	0	1	2	20	71	131		
	CDH23.3	7202	1814	1701	1230	379	66	1	1	4	30	179	607	1190		
	CDH26.7	3077	841	792	511	104	8	0	0	0	4	49	236	532		
(14) Wind	WSAvg	2.9	3.7	3.6	3.2	2.5	2.1	2.2	2.3	2.4	2.8	3.1	3.5	3.7		
(15) Precipitation	PrecAvg	820	7	18	18	45	112	177	169	114	75	52	23	11		
	PrecMax	1042	51	131	72	98	222	294	272	194	184	126	73	32		
	PrecMin	560	0	0	0	0	14	68	88	18	18	1	3	0		
	PrecStd	136	11	27	19	28	52	64	46	39	38	27	18	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.2	38.4	37.4	32.5	27.7	23.0	21.8	23.5	26.6	31.2	35.5	38.1		
		MCWB	20.0	21.2	20.3	18.3	16.0	14.0	13.8	14.2	15.5	17.4	18.2	19.7		
	2%	DB	35.8	35.7	34.0	29.5	25.2	21.2	19.9	21.2	23.4	27.7	32.1	34.9		
		MCWB	20.1	20.6	19.8	17.6	15.8	14.2	13.9	14.4	15.0	16.6	17.4	19.1		
	5%	DB	33.6	33.7	31.6	27.3	23.1	19.9	18.7	19.6	21.3	25.0	29.2	32.2		
		MCWB	20.3	20.3	19.3	17.5	15.6	14.0	13.5	13.9	14.6	15.9	17.4	19.0		
	10%	DB	31.4	31.6	29.4	25.3	21.6	18.7	17.5	18.2	19.6	22.8	26.8	29.5		
		MCWB	19.8	20.0	18.8	17.4	15.3	13.8	13.2	13.5	14.1	15.5	17.0	18.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.1	24.2	22.7	20.9	19.3	17.5	16.8	17.3	17.5	19.1	20.5	22.0		
		MCDB	32.2	32.3	29.6	25.1	22.0	19.7	18.6	19.7	21.7	26.3	28.4	32.5		
	2%	WB	22.6	22.9	21.6	20.0	18.2	16.4	15.6	16.0	16.5	18.0	19.5	21.0		
		MCDB	30.2	30.0	28.3	24.6	21.4	18.8	17.7	18.8	20.6	24.2	27.0	30.0		
	5%	WB	21.6	21.8	20.8	19.2	17.3	15.5	14.7	15.1	15.6	17.1	18.7	20.1		
		MCDB	29.5	30.2	27.6	24.1	20.7	18.1	17.0	18.0	19.7	22.7	25.7	28.7		
	10%	WB	20.7	20.9	20.1	18.4	16.3	14.6	13.9	14.2	14.7	16.3	17.9	19.3		
		MCDB	28.4	29.0	26.7	23.4	20.1	17.3	16.4	17.2	18.6	21.4	24.6	26.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	11.9	11.6	11.0	11.0	10.1	9.5	9.7	9.8	10.5	10.8	11.0		
		MCDBR	15.6	15.2	14.9	13.8	13.7	12.0	10.9	12.4	13.1	14.3	15.6	15.7		
		MCWBR	5.4	5.2	5.0	5.5	6.4	6.4	6.1	6.6	6.3	6.2	5.7	5.4		
	5% WB	MCDBR	12.1	12.4	11.4	10.3	9.5	9.0	8.0	9.1	10.3	11.4	12.2	12.7		
		MCWBR	4.9	4.9	4.5	5.0	5.6	5.9	5.5	5.7	5.9	5.5	5.1	5.0		
(40) Clear-Sky Solar Irradiance	taub		0.340	0.341	0.325	0.318	0.308	0.301	0.300	0.308	0.327	0.327	0.329	0.329		
	taud		2.579	2.593	2.629	2.635	2.635	2.633	2.634	2.605	2.527	2.545	2.566	2.585		
	Ebn at Noon		999	979	961	917	877	858	877	915	943	981	1004	1013		
	Edh at Noon		106	101	91	81	73	69	72	82	99	105	106	106		
(44) All-Sky Solar Radiation	RadAvg		8.19	7.38	6.10	4.45	3.33	2.71	2.88	3.74	4.99	6.50	7.75	8.39		
	RadStd		0.38	0.25	0.30	0.26	0.22	0.19	0.18	0.18	0.29	0.28	0.33	0.34		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (20 neighbors)	N/A	N/A	N/A	+0.50	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+0.54	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page